

ARYAN SHARMA

Voice AI & Real-Time Systems Engineer

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SUMMARY

Voice AI and real-time systems engineer. Built V■YU, a production-grade voice-first RAG system that answers in 0.8 ms P50 WebSocket round-trips against a 50–100 ms requirement — with speculative retrieval while the user is still speaking, barge-in cancellation, and per-stage latency telemetry on every hop. Deep experience with STT/TTS pipelines (Sarvam), LPU-accelerated LLM streaming, and the network-level fixes (TCP_NODELAY, Nagle, delayed ACK) that make voice feel instant.

TECHNICAL SKILLS

- Speech: Sarvam AI (saaras:v2 STT, bulbul:v2 TTS) · Web Speech API · buffered mic capture · partial transcripts · TTS streaming (WebSocket WAV)
- Real-Time: WebSockets · TCP_NODELAY/Nagle profiling · P50/P70/P100 latency telemetry · speculative retrieval · barge-in / generation cancellation · LRU caches
- RAG: FAISS (IndexFlatIP) · 9,000+ in-RAM vectors · TF-IDF (512-dim) · parent-child chunking · citation grounding
- LLM Serving: Groq LPU (Llama3-8b) streaming · token-budget control · deterministic guardrails (<1 ms regex)
- Languages: Python · TypeScript · Rust · Go · SQL · Infra: FastAPI, Uvicorn, Next.js, Docker

EXPERIENCE

Cinntra Infotech — Systems Engineer · 2026 – Present

- Built CIRA, a RAG agent translating natural-language requests into audited SAP HANA queries and write operations, giving non-technical users direct access to enterprise data
- Built the point-of-sale system backend: transaction recording, merchant reconciliation, and offline synchronization

Niswa Engineering — Technical Consultant & Automation Engineer · 2024 – 2026

- Built asynchronous AI agents (TypeScript) that automated 80%+ of a content generation and scheduling pipeline
- Built the company website in Astro (95+ Google Lighthouse score); set up cloud infrastructure, API integrations, and event-driven automation workflows

Sudarshana Foundation — Full-Stack & Automation Engineer · Contract

- Designed and built the organization's first centralized web platform, digitizing paper-based workflows into software
- Built database systems handling 5,000+ monthly transactions

PROJECTS

V■YU — Ultra-Low-Latency Voice-Enabled RAG System

- Beat a strict 50–100 ms end-to-end voice-answer requirement by ~99%: measured P50 0.8 ms, P70 1.5 ms, worst-case 1.6 ms client→server WebSocket round-trips via a live 10-query audit harness.
- Root-caused and fixed a Nagle + delayed-ACK pathology — uvicorn's websockets path never set TCP_NODELAY, adding ~42 ms to every round trip — cutting WS latency from ~44 ms to ~1.5 ms (96% reduction); added per-stage telemetry (guardrail 0.005 ms → embed 0.51 ms → FAISS 0.017 ms → parent 0.009 ms → grounding 0.005 ms; 0.67 ms P50 retrieval).
- Designed the voice UX pipeline: Sarvam STT on buffered mic audio with browser partials for live transcripts, speculative background retrieval triggered by partials (candidates in RAM before speech ends), Groq Llama 3 streaming tokens, Sarvam TTS (speaker-selectable) streaming WAV back over WebSocket — with engine attribution (SARVAM vs BROWSER voice) in the answer meta.
- Tech: Python, FastAPI, WebSockets, FAISS, Groq LPU, Sarvam AI, Next.js 16 — github.com/Aryan-Protein-Vala/vayu-voice-rag

MYLO OS — Voice-First Desktop Operator (Tauri 2 / Rust)

- Built the voice-trigger layer for a desktop OS agent: hold a global hotkey, speak the task, and a ghost cursor executes it across the OS — screen understanding via zero-latency Windows Graphics Capture / macOS Core Graphics at ~35 MB RSS.
- Tech: Rust, Tauri 2, Windows Graphics Capture, Core Graphics — github.com/Aryan-Protein-Vala/MyloOS

CLAIM/05 — Real-Time Multiplayer Sync Engine (Go + Redis)

- Built a Go WebSocket engine (hub pattern, concurrent channel hubs) syncing a 5,000×5,000 (25M-cell) live battlefield: Redis SETNX atomic cell locks guarantee conflict-free ownership, and Pub/Sub delta fan-out reaches every connected client in sub-millisecond — the same real-time fabric voice UIs need.
- Tech: Go, Gorilla WebSockets, Redis Pub/Sub, Next.js 16 — github.com/Aryan-Protein-Vala/Grid-Game · live: multiplayer-grid-game-theta.vercel.app

B1 Copilot — Streaming Analytics Copilot (SAP)

- Built streaming token delivery of LLM-generated SQL answers over WebSockets, interleaved with live table and chart rendering as each query stage completes — partial results visible before generation finishes.
- Tech: Python, FastAPI, WebSockets, Next.js 16, SAP HANA — github.com/Aryan-Protein-Vala/Cira-RAG-agent

CERTIFICATIONS

- Frontend Development Certification — Sheryians Coding School
- Data Science & AI/ML Bootcamp Certification — Udemy